https://rahattou.net

$\underline{P} = (E, \vec{p})$ $\quad\quad$ $\underline{x} = (t, \vec{x})$ $\quad\quad$ $g^{\mu\nu} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix}$

$E^2 = p^2 + m^2$ $\quad\quad$ $|\underline{P}|^2 = E^2 - |\vec{P}|^2$

Natural units: $\hbar = c = 1$

$\hbar c = 197$ MeV x fm $\quad\quad$ $1 \text{ fm} = 10^{-15}$ m

$\Delta p . \Delta x \simeq 1$ $\quad\quad$ $1 \text{ fm} \simeq \frac{1}{200} \text{ MeV}^{-1}$

Δx $\quad\quad$ $\Delta V = 1V$

$E = q\Delta V = e\Delta V = 1 eV$

$= 1.6 \times 10^{-19}$ J

$c = 3 . 10^8$ m/s

$e = 1.6 \times 10^{-19}$ C

$\hbar = \frac{h}{2\pi}$ $\quad\quad$ $E = h\nu$ $\quad\quad$ $[h] = [E][T]$

Two ingredients:

1/ collisions

2/ Decays

Decay: unstable particles decay

$a \to b + c$ $\quad\quad$ $n \to p\, e^- \bar{\nu}_e$

Free state/particle: $\quad$ $\psi(x,t) = \underbrace{\psi(x)\, e^{-iEt}}_{H_0}$

$|\psi(x)|^2$

$H = H_0 + H_I$

↳ free hamiltonian

$\frac{p^2}{2m}$

$i \frac{d\psi}{dt} = H\psi$

Perturbation theory at least 2nd order

$$E' = E_0 + \Delta E_1 - i\,\underbrace{\Delta E_2/2}_{\text{Variation from 2nd order.}} + O(3)$$

$E_0 \downarrow$ free energy

$$\psi(x,t) = \psi(x)\, e^{-i(E_0 + \Delta E_1)t}\, e^{-i\left(-i\frac{\Delta E_2}{2}\right)t}$$

$$= \psi(x)\, e^{-iE't}\, e^{-\frac{\Gamma}{2}t}$$

$$|\psi(x,t)|^2 = e^{-\Gamma t}$$

$$\lim_{t\to\infty} |\psi(x,t)|^2 = 0 \qquad \text{Decay}$$

$a \to b + c$ two body decay.

$a \to b + c + d$ 3-body decay

$a \to b + c + d + \dots + z$ n-body decay

$n \to p + e^- + \bar{\nu}_e$ β^--decay

Atomic # / electric charge $\leftarrow$ ${}^{A}_{Z}N \to {}^{A}_{Z+1}N + e^- + \bar{\nu}_e$

~~$p \to n + e^+ + \nu_e$~~ β^+

${}^{A}_{Z}N \to {}^{A}_{Z-1}N + e^+ + \nu_e$

${}^{A}_{Z}N \to {}^{A-4}_{Z-2}N + \alpha$ α decay.

$\pi^0 \to \gamma\gamma$

$K^0 \to \pi^+\pi^-$

$B^0 \to D^{*+}\, e^-\, \bar{\nu}_e$

$a \to b + c$

$Q = m_a - m_b - m_c \geq 0$ decay can happen

$b \leftarrow \overset{a}{\bullet} \rightarrow c$

rest frame of a: $E_a = E_b + E_c$

$\vec{P_a} = \emptyset$

$m_a = m_b + \underbrace{K_b}_{\text{kinetic energy.}} + m_c + K_c$

$E = \sqrt{p^2 + m^2} = m + \underbrace{K}_{\text{kinetic energy.}}$

N_0 particles unstable: $N(t) = N_0 \, e^{-t\Gamma} = e^{-t/\tau}$

Exponential decay rate

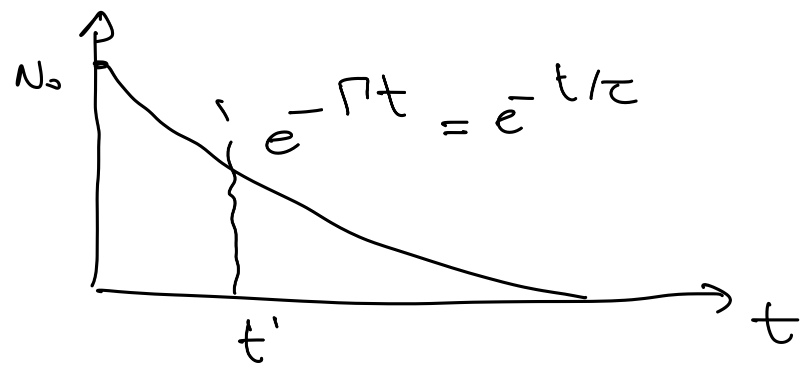
$\frac{N(t)}{N_0} \leq 1$ $[\Gamma] = T^{-1} = E$

Γ: decay rate measured in MeV

$\tau = \frac{1}{\Gamma}$ mean lifetime

$$\tau = \langle t \rangle = \frac{\int_0^\infty t \, e^{-t\Gamma} dt}{\int_0^\infty e^{-t\Gamma} dt} = \frac{1}{\Gamma}$$

$\mu^- \to e^- \bar{\nu}_e \nu_\mu$ $\tau_\mu \simeq 2.2 \; 10^{-6}$ s.



Fermi's golden Rule: $\qquad i \longrightarrow f$

$$\Gamma_{i\to f} = 2\pi\, |M_{fi}|^2\, \rho(E)\Big|_{E_f=E_i}$$

probability per unit time

$|i\rangle = |a\rangle \qquad a \longrightarrow b+c$

$|f\rangle = |b+c\rangle$

$$M_{fi} = \langle f|\, H_I\, |i\rangle$$

$\rho(E)$: phase space.

$a \to b+c \qquad a \to f+g$

$K^0 \to \pi^+\pi^- \qquad K^0 \to \pi^+\pi^-\pi^0$

$\pi^+ \to e^+\nu_e \qquad \pi^+ \to \mu^+\nu_\mu$

larger Q $\Longrightarrow$ larger phase space $\rho(E)$

$b \longleftarrow \overset{a}{\circ} \longrightarrow c$

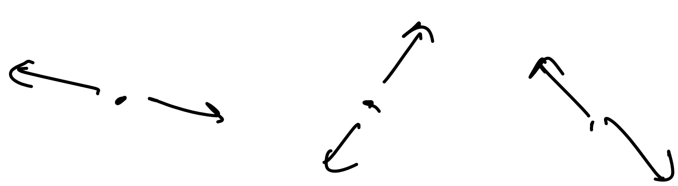
energy: $\quad m_a = E_a + E_b = \sqrt{p_b^2+m_b^2} + \sqrt{p_c^2+m_c^2}$

momentum: $\quad \vec{0} = \vec{p}_b + \vec{p}_c$

$$\Rightarrow \vec{p}_b = -\vec{p}_c = \vec{p}$$

$$m_a = \sqrt{p^2+m_b^2} + \sqrt{p^2+m_c^2}$$

$$p = \frac{m_a^2 - m_b^2 - m_c^2}{2m_a} \qquad Q = m_a - m_b - m_c$$



Relativistic Golden Rule

$$1 \longrightarrow 2 + 3 + 4 + \cdots + n$$

$|i\rangle = |1\rangle$

$|f\rangle = |2+3+4+\cdots n\rangle$

$$|M_{fi}| = |\langle f | H_I | i \rangle|$$

conservation of $\underline{P}(E, \vec{p})$

$$\Gamma_{i\to f} = \frac{S}{2m_1} \int |M_{fi}|^2 (2\pi)^4 \, \delta^4(\underline{P}_1 - \underline{P}_2 - \underline{P}_3 - \cdots - \underline{P}_n) \times$$

$$\prod_{j=2}^{n} (2\pi)\, \underbrace{\delta(\underline{P}_j^2 - m_j^2)}_{\text{j-th particle on-shell}}\, \theta(E_j) \, \frac{d^4 \underline{P}_j}{(2\pi)^4}$$

$E^2 = p^2 + m^2$

$E = \pm\sqrt{p^2 + m^2}$

S factor: $\quad a \to b + b + c + c + c$

$$S = \frac{1}{2!} \frac{1}{3!}$$

$K^0 \to \pi^0 \pi^0 \pi^0 \quad S = \frac{1}{3!}$

$K^0 \to \pi^+ \pi^- \pi^0 \quad S = 1$

$$\beta = \frac{v}{c} = \frac{|\vec{p}|}{E} \qquad \gamma = \frac{E}{m} \qquad \beta\gamma = \frac{|\vec{p}|}{m}$$